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United States Patent Application Publication

20250283651

Kind Code

A1

Publication Date

September 11, 2025

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COMPRESSOR CHASSIS AND COOLING APPLIANCE HAVING THE SAME

Abstract

A cooling appliance includes a main body in which a compressor chamber is defined, a chassis which is in the compressor chamber, the chassis defining a fastening portion to which a fastening member is screw-coupled to the chassis, and a support portion which is extended from the fastening portion and connects the fastening portion to a portion of the chassis which is adjacent to the support portion, and a compressor of a cool air generator which is in the compressor chamber, penetrated by the fastening member and detachably attached to the chassis by the fastening member.

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Family ID: 1000008640948

Appl. No.: 19/217034

Filed: May 23, 2025

Foreign Application Priority Data

KR

10-2022-0165870

Dec. 01, 2022

Related U.S. Application Data

parent WO continuation PCT/KR2023/015683 20231012 PENDING child US 19217034

Publication Classification

Int. Cl.: F25D19/00 (20060101); B23P15/00 (20060101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of International Application No. PCT/KR2023/015683 designating the United States, filed on Oct. 12, 2023, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent Application No. 10-2022-0165870 filed on Dec. 1, 2022 in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

(1) Field

[0002] The disclosure relates to a refrigerator as a cooling appliance having a compressor attachment portion to which a compressor is bolt-fastened.

(2) Description of the Related Art

[0003] In general, a refrigerator is a home appliance including a storage chamber for storing food, and a cold air generation unit for supplying cold air to the storage chamber through a refrigeration cycle to keep the food fresh.

[0004] The cold air generation unit drives the refrigeration cycle to supply cold air to the storage chamber. The cold air generation unit includes a compressor which compresses a gaseous refrigerant to a high-temperature and high-pressure state, a condenser which condenses the compressed refrigerant into a liquid phase, an expansion device which expands the condensed refrigerant, and an evaporator which evaporates the liquid-state refrigerant to generate cold air.

SUMMARY

[0005] As a result of condensing the gaseous refrigerant which is compressed into the high-temperature and high-pressure state, into a liquid state, the condenser radiates heat to the outside. The heat radiation of the condenser raises the temperature of air around the condenser. Therefore, the cold air generation unit further includes a blower fan to exhaust the high-temperature air around the condenser to the outside and to continuously draw outside air into the condenser.

[0006] The disclosure provides a compressor attachment portion enabling a compressor to be bolt-fastened directly to a chassis without a separate connection member.

[0007] According to an embodiment of the disclosure, there is provided a refrigerator including a main body, a machinery chamber provided in a lower portion of the main body, a compressor disposed in the machinery chamber, a chassis supporting the compressor, and a compressor attachment portion provided in the chassis and allowing the compressor to be detachably attached thereto, in which the compressor attachment portion includes a support portion integrally formed with the chassis, and a fastening portion extending from the support portion in a burring shape and screw-coupled to a bolt penetrating a partial portion of the compressor.

[0008] The support portion may have the same thickness as the chassis.

[0009] The thickness of the support portion may be about 0.8 millimeter (mm) to about 1.0 mm. The fastening portion has a larger thickness than the support portion. The thickness of the fastening portion may be about 1.15 millimeters (mm) to about 1.25 mm.

[0010] The thickness of the support portion may be about 1.2 mm to about 1.4 mm. The fastening portion may have a smaller thickness than the support portion. The thickness of the fastening portion may be about 1.1 mm to about 1.3 mm.

[0011] The fastening portion may have a height of 4.0 about mm to 4.5 about mm from an upper surface of the support portion.

[0012] According to another embodiment of the disclosure, there is provided a molding method for a compressor attachment portion to which a compressor provided in a refrigerator is attached. The molding method includes a drawing process for forming a molded portion by deforming a portion of a chassis connected to the compressor to be convex using a first drawing punch, a re-drawing process for increasing a height of the molded portion and decreasing a diameter of the molded portion using a second drawing punch having a smaller diameter than the first drawing punch, a piercing process for forming a through hole in an upper end of the molded portion by perforating the molded portion using a piercing punch, a first forging process for forming a fastening portion deformed to have a burring shape from the molded portion by penetrating the molded portion using a burring punch, and striking an upper end of the fastening portion using a first forging punch to thicken the fastening portion to have a first thickness, and a second forging process for striking the upper end of the fastening portion using a second forging punch to thicken the fastening portion to have a second thickness larger than the first thickness and larger than a thickness of a support portion extending to a lower end of the fastening portion.

[0013] The re-drawing process may include a first re-drawing process for deforming the fastening portion to have a first height and a first diameter, and a second re-drawing process for deforming the fastening portion to have a second height larger than the first height and a second diameter smaller than the first diameter.

[0014] The molding method may further include a tapping process for forming a screw thread on an inner circumferential surface of the fastening portion using a cutting tool.

[0015] According to another embodiment of the disclosure, there is provided a molding method for a compressor attachment portion to which a compressor provided in a refrigerator is attached. The molding method includes a drawing process for forming a molded portion by deforming a portion of a chassis connected to the compressor to be convex using a first drawing punch, a piercing process for forming a through hole in an upper end of the molded portion by perforating the molded portion using a piercing punch, a re-drawing process for increasing a height of the molded portion and decreasing a diameter of the molded portion using a second drawing punch having a smaller diameter than the first drawing punch, a molding process for decreasing the diameter of the molded portion using a molding punch having a smaller diameter than the second drawing punch, and a burring process for forming a fastening portion by penetrating the molded portion using a burring punch to deform the molded portion to have a burring shape, in which the fastening portion is formed to have a thickness smaller than a thickness of a support portion extending to a lower end of the fastening portion.

[0016] The re-drawing process may include a plurality of re-drawing processes for deforming the fastening portion to gradually increase a height of the fastening portion and gradually decrease a diameter of the fastening portion.

[0017] The molding method may further include a tapping process for forming a screw thread on an inner circumferential surface of the fastening portion using a cutting tool.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] The above and/or other aspects of the disclosure will be more apparent by describing embodiments of the disclosure with reference to the accompanying drawings, in which:

[0019] FIG. 1 is a perspective view illustrating a front side of a refrigerator according to an embodiment of the disclosure.

[0020] FIG. 2 is an exploded perspective view illustrating a rear side of the refrigerator according to an embodiment of the disclosure.

[0021] FIG. 3 is an exploded perspective view illustrating a compressor attachment portion

provided in the refrigerator according to an embodiment of the disclosure.

[0022] FIG. 4 is a perspective view illustrating a chassis in which the compressor attachment portion provided in the refrigerator is provided according to an embodiment of the disclosure.

[0023] FIG. 5 is a cross-sectional view taken along line A-A illustrated in FIG. 4.

[0024] FIG. 6 is a flow chart illustrating a method of providing the compressor attachment portion according to an embodiment of the disclosure.

[0025] FIGS. 7A to 7G are cross-sectional diagrams illustrating the method of providing the compressor attachment portion according to an embodiment of the disclosure.

[0026] FIG. 8 is a flow chart illustrating a method of providing the compressor attachment portion according to an embodiment of the disclosure.

[0027] FIGS. 9A to 9H are cross-sectional diagrams illustrating the method of providing the compressor attachment portion according to an embodiment of the disclosure.

DETAILED DESCRIPTION

[0028] It should be understood that embodiments to be described below are exemplarily provided to help the understanding of the disclosure, and the disclosure may be modified in various ways, unlike the embodiments described herein. However, in the following description of the disclosure, if it is determined that a detail description of a related known function or component may unnecessarily obscure the gist of the disclosure, the detailed description and concrete illustration thereof will be omitted. Further, the accompanying drawings are not necessarily illustrated to actual scale but dimensions of some components may be exaggerated to help the understanding of the disclosure.

[0029] It will be understood that when an element is referred to as being related to another element such as being “on” another element, it can be directly on the other element or intervening elements may be present therebetween. In contrast, when an element is referred to as being related to another element such as being “directly on” another element, there are no intervening elements present.

[0030] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used herein, “a,” “an,” “the,” and “at least one” do not denote a limitation of quantity, and are intended to include both the singular and plural, unless the context clearly indicates otherwise. Thus, reference to “an” element in a claim followed by reference to “the” element is inclusive of one element and a plurality of the elements. For example, “an element” has the same meaning as “at least one element,” unless the context clearly indicates otherwise. “At least one” is not to be construed as limiting “a” or “an.” “Or” means “and/or.” As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” or “includes” and/or “including” when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof.

[0031] The terms “first,” “second” and the like may be used to describe various components, but the components should not be limited by these terms. These terms are used only for the purpose of distinguishing one component from another component. For example, a first component may be referred to as a second component, and similarly, the second component may also be referred to as the first component without departing from the scope of the disclosure.

[0032] Unless otherwise defined, the terms used in the embodiments of the disclosure may be interpreted as having meanings commonly known to those having ordinary skill in the art.

[0033] The terms such as “front end,” “rear end,” “upper portion,” “lower portion,” “upper end,” and “lower end” used in the disclosure are defined based on the drawings, and the shape and position of each component are not limited by the terms.

[0034] The expression “same” used in the disclosure may not only mean a complete match, but also include a difference to such an extent that a processing error range is considered.

[0035] Meanwhile, technical elements and areas in the drawings are illustrated schematically. Accordingly, the technical idea of the disclosure is not limited by the relative sizes or intervals illustrated in the accompanying drawings. In describing the disclosure, if it is determined that a detail description of a related known function or component may unnecessarily obscure the gist of the disclosure, the detailed description thereof will be simplified or omitted.

[0036] “About” or “approximately” as used herein is inclusive of the stated value and means within an acceptable range of deviation for the particular value as determined by one of ordinary skill in the art, considering the measurement in question and the error associated with measurement of the particular quantity (i.e., the limitations of the measurement system). For example, “about” can mean within one or more standard deviations, or within $\pm 30\%$, 20% , 10% or 5% of the stated value.

[0037] Hereinafter, a refrigerator **1** as a cooling appliance according to an embodiment of the disclosure will be described in detail with reference to the accompanying drawings so that it can be easily implemented by those having ordinary skill in the art to which the disclosure pertains.

[0038] FIG. **1** is a perspective view illustrating a front side of a refrigerator **1** according to an embodiment of the disclosure. FIG. **1** shows the cooling appliance being open at the front side.

[0039] Referring to FIG. **1**, a refrigerator **1** includes a main body **10**, a storage compartment **20**, and a door **30**.

[0040] The main body **10** includes an outer case **11** and an inner case **13**. The outer case **11** forms an exterior appearance of the main body **10**. The outer case **11** may be made of or include a metal material having excellent durability and aesthetics. The inner case **13** is positioned inside the outer case **11**. The inner case **13** forms an exterior appearance of the storage compartment **20**. The inner case **13** may be integrally injection-molded with a plastic material. An insulating material may be foamed (or provided) between the inner case **13** and the outer case **11** to prevent cold air from flowing out of the storage compartment **20**.

[0041] The storage compartment **20** is exposed to outside the main body **10** being open at the front side of the cooling appliance so that food can be taken in and out. Here, the main body **10** may define an opening through which the storage compartment **20** is accessible from the front side of the cooling appliance. According to an embodiment, the storage compartment **20** may be divided into a plurality of storage compartments (or sub-compartments) by a partition wall **17**. The partition wall **17** may be disposed within the inner case **13**. The storage compartment **20** divided into a plurality of storage compartments by the partition wall **17** may include an upper storage compartment **21** and a lower storage compartment **23**. Here, the main body **10** may define openings through which the sub-compartments are respectively accessible from outside the cooling appliance, through the open front side thereof. The upper storage compartment **21** and the lower storage compartment **23** may be separated by a first partition wall **17a**. The lower storage compartment **23** may be divided into a left storage compartment **23a** and a right storage compartment **23b** by a second partition wall **17b**.

[0042] The storage compartment **20** may include a refrigerating compartment and a freezing compartment. Depending on the type of refrigerator **1**, the upper storage compartment **21** may be provided as a refrigerating compartment and the lower storage compartment **23** may be provided as a freezing compartment. Alternatively, the upper storage compartment **21** may be provided as a freezing compartment and the lower storage compartment **23** may be provided as a refrigerating compartment. The freezing compartment and the refrigerating compartment may be thermally insulated by the first partition wall **17a**. The freezing compartment may be maintained at about -20 degrees Celsius ($^{\circ}$ C.), and the refrigerating compartment may be maintained at about 3° C. However, the temperature of the freezing compartment and refrigerating compartment may be changed according to various conditions such as the external temperature and the installation location of the cooling appliance.

[0043] A shelf **25** and a storage container **27** may be installed inside the storage compartment **20**.

The shelf **25** is provided to support food stored in the storage compartment **20**. A plurality of shelves **25** may be provided in each storage compartment **20**. The shelf **25** may be provided to be detachable from and attachable to the storage compartment **20**. The storage container **27** may be provided in a box shape. The storage container **27** may be provided to store food in a sealed inner space. The storage container **27** may be detachable from and attachable to the storage compartment **20**.

[0044] The storage compartment **20** is opened and closed by a position of the door **30** relative to the main body **10**. The door **30** is pivotably connected to the main body **10** to open and close the open front side of the storage compartment **20**. The upper storage compartment **21** and the lower storage compartment **23** are opened and closed by an upper door **31** and a lower door **33**, respectively, which are each pivotably connected to the main body **10**. Each of the upper door **31** and the lower door **33** may be provided in a double-door type. Here, each of the upper door **31** and the lower door **33** may be two sub-doors respectively connected to opposing sides of the main body **10** and respectively functioning to open and close a portion of the storage compartment **20**. A plurality of door guards **35** in which food and the like may be stored, may be provided on the back (e.g., inner) sides of the upper door **31** and the lower door **33**. These door guards **35** may also be detachable from and attachable to respective sub-doors.

[0045] FIG. **2** is an exploded perspective view illustrating a machinery chamber and a cold air generation unit from the rear side of the refrigerator **1** according to an embodiment of the disclosure.

[0046] The machinery chamber **40** may be a volume or space within the main body **10** which is accessible from the rear side of the cooling appliance, such as through a chamber opening defined in the main body **10**. The machinery chamber **40** may include a rear cover **42** which separates the volume of the machinery chamber **40** from outside the cooling appliance when installed to the main body **10**. The rear cover **42** may be provided to open and close the machinery chamber **40** at the rear side of the main body **10** which is opposite to the front side of the cooling appliance along a first direction, e.g., a front-back direction.

[0047] The rear cover **42** may have an inlet port **42a** through which air is brought into the machinery chamber **40** from outside the main body **10**, and an outlet port **42b** through which air inside the machinery chamber **40** is discharged to the outside (e.g., the outside of the main body **10**). The inlet port **42a** may be provided in plural including a plurality of inlet ports **42a** and the outlet port **42b** may be provided in plural including a plurality of outlet ports **42b**. The inlet port **42a** and the outlet port **42b** may be provided at different positions along the rear cover **42**. In an embodiment, the various air ports may be defined in a body of the rear cover **42**, such as being enclosed openings extended completely through a thickness of the cover body to allow airflow between the machinery chamber **40** and the outside. The group of air inlet ports and the group of air outlet ports are shown in FIG. **2** to be adjacent to each other along a second direction, e.g., a left-right direction.

[0048] The refrigerator **1** according to an embodiment of the disclosure may further include a cold air generation unit **50** which generates and supplies cold air to the storage compartment **20**. The cold air generation unit **50** may include a compressor **51**, a condenser **60**, an expansion valve (not shown), and an evaporator (not shown). The cold air generation unit **50** may operate a refrigeration cycle of the cooling appliance by using the compressor **51**, the condenser **60**, the expansion valve and the evaporator, thereby generating cold air.

[0049] The compressor **51** compresses a refrigerant to a high-temperature and high-pressure state. The compressor **51** may be supplied with electric energy from the outside to compress a gaseous refrigerant to a high-temperature and high-pressure state using a rotational force of an electric motor or the like. The compressor **51** may be connected (e.g., fluidly, electrically, etc.) to the condenser **60** and operate to move the compressed refrigerant to the condenser **60**. The compressor **51** may be positioned inside the machinery chamber **40** (e.g., a compressor chamber). The

compressor **51** compresses the refrigerant and pushes the compressed refrigerant into the condenser **60** to operate a refrigeration cycle including processes of compression, condensation, expansion, and evaporation. Therefore, when the compressor **51** is in operation, cold air generated in the evaporator is supplied to the storage compartment **20**.

[0050] The condenser **60** condenses the high-temperature and high-pressure refrigerant which is compressed by the compressor **51**. The condenser **60** dissipates heat generated while condensing the refrigerant. The refrigerant which is condensed while passing through the condenser **60** moves to the expansion valve. That is, the condenser **60** and the expansion valve may be connected to each other. The refrigerant which is condensed by the condenser **60** becomes a low-temperature and low-pressure liquid state while passing through the expansion valve. The liquid-state refrigerant passes through the expansion valve and moves to the evaporator. That is, the expansion valve and the evaporator may be connected to each other.

[0051] The evaporator evaporates the low-temperature and low-pressure liquid refrigerant which has passed through the expansion valve. While the liquid refrigerant is evaporated by the evaporator, the refrigerant exchanges heat with surrounding gas. While the liquid refrigerant is evaporated, the refrigerant absorbs latent heat from the surroundings, thereby cooling the gas surrounding the evaporator and generating cold air. The completely evaporated refrigerant is supplied back to the compressor **51**, so that the cooling cycle continues.

[0052] Some components of the cold air generation unit **50** as a cool air generator may be positioned inside the machinery chamber **40**. According to an embodiment of the disclosure, the compressor **51**, the condenser **60**, and a blower fan **53** may be positioned inside the machinery chamber **40**.

[0053] FIG. **3** is an exploded perspective view illustrating a compressor attachment portion **100** of the refrigerator **1** according to an embodiment of the disclosure.

[0054] Referring to FIG. **3**, the compressor **51** may be fixed to a chassis **70** disposed at the bottom of the machinery chamber **40**, via a plurality of bolts **80**. The compressor **51** may have a support piece **51a** formed (or provided) at each of the front side and the rear side of the lower portion thereof. The support piece **51a** may be formed to protrude from both sides of the compressor **51**, such as protruding further from a body of the compressor **51** to allow attachment of the body to the chassis **70**.

[0055] A damper **55** may be coupled to each of both ends of a support piece **51a**. The damper **55** may have a through hole **55a** defined therein and formed along a length direction. The bolt **80** may be inserted into the through hole **55a** of the damper **55**. A leading end portion of the bolt **80** inserted into the through hole **55a** of the damper **55** may be screwed to a fastening portion **115** of a compressor attachment portion **100** of the chassis **70**. Here, the fastening portion **115** may correspond to a location of the through hole **55a**, to allow attachment of the compressor **51** to the chassis **70**.

[0056] The damper **55** may be made of an elastic material to absorb vibrations. The damper **55** may be disposed between the support piece **51a** of the compressor **51** and the chassis **70** along a third direction (e.g., a top-down direction or thickness direction). Two dampers **55** may be respectively disposed on both sides of the front of the compressor **51**, and two dampers **55** may be respectively disposed on both sides of the rear of the compressor **51**. The plurality of dampers **55** can absorb vibrations generated when the compressor **51** is driven and reduce noise caused by the vibrations.

[0057] The compressor attachment portion **100** according to an embodiment of the disclosure may be provided in the chassis **70**. Here, a body of the chassis **70** may define the compressor attachment portion **100**. The compressor attachment portion **100** may be provided in a number corresponding to the attachment points at which the compressor **51** and the chassis **70** are connected to each other by the bolts **80**. For example, as shown in FIG. **3**, four compressor attachment portions **100** may be provided in the chassis **70** to correspond to four bolts **80** used for a four-point connection of the compressor **51** to the chassis **70**.

[0058] FIG. 4 is a plan view illustrating a chassis **70** in which the compressor attachment portion **100** is provided according to an embodiment of the disclosure. FIG. 5 is a cross-sectional view taken along line A-A illustrated in FIG. 4.

[0059] Referring to FIG. 4, the compressor attachment portion **100** may be provided in or defined by portions of the chassis **70**. For example, the compressor attachment portion **100** may be integrally formed with a remainder of the chassis **70** through a sheet forging and thickening method. The compressor attachment portion **100** may include a support portion **110** and a fastening portion **115** which protrudes from the support portion **110**.

[0060] The chassis **70** may have a plate portion along which the compressor attachment portion **100** is defined. The plate portion may have an upper surface **71** facing the storage compartment **20**. The upper surface **71** may be disposed in a plane defined by the first and second directions crossing each other.

[0061] An upper surface **110a** (see FIG. 5) of the support portion **110** may be disposed at a height lower than an upper surface **71** of the chassis **70**. That is, the compressor attachment portion **100** may be recessed from the plate portion of the chassis **70**. In this case, the compressor attachment portion **100** may have a groove shape, and the support portion **110** may form a bottom of the groove shape. In a case where the compressor attachment portion **100** has a groove shape, a lower end of the damper **55** disposed on the compressor attachment portion **100** may be inserted into the recess of the compressor attachment portion **100**. That is, a portion of the damper **55** may be accommodated in the depth of the compressor attachment portion **100**.

[0062] The compressor attachment portion **100** is not limited to the structure described above. For example, the upper surface **110a** of the support portion **110** may be positioned at the same height as the upper surface **71** of the chassis **70**. In this case, the compressor attachment portion **100** may be provided on a same plane as the upper surface **71** of the chassis **70** (e.g., coplanar with each other).

[0063] Referring to FIG. 5, the fastening portion **115** may protrude from the support portion **110** to a predetermined height. The fastening portion **115** may have a fastening hole **116** defined therein into which the bolt **80** (see FIG. 3) is detachably coupled. A screw thread **117** may be formed on an inner circumferential surface of the fastening hole **116**.

[0064] Along a depth direction of the compressor attachment portion **100** (or a thickness direction of the chassis **70**), the fastening portion **115** may have a height **H** for securing a length of the screw thread **117** sufficient to stably fasten the bolt **80** into the fastening hole **116**. Here, the height **H** of the fastening portion **115** may correspond to a length from the upper surface **110a** of the support portion **110** to an upper end **115a** (e.g., distal end) of the fastening portion **115**.

[0065] In an embodiment, an upper surface of the support portion **110** may face the compressor **51**, and the fastening portion **115** may protrude from the upper surface. For example, the height **H** of the fastening portion **115** may be about 4.0 millimeters (mm) to about 4.5 mm in consideration of the fastening torque (e.g., about 40 kilogram-force (kgf) or more) between the fastening portion **115** and the bolt **80**. If the height **H** of the fastening portion **115** is less than about 4.0 mm, it is difficult to secure a stable fastening torque between the fastening portion **115** and the bolt **80** because the length of the screw thread **117** which can be processed on the inner circumferential surface of the fastening hole **116** is short. If the height **H** of the fastening portion **115** exceeds about 4.5 mm, assemblability may deteriorate as the lower end of the damper **55** interferes with the upper end portion of the fastening portion **115**.

[0066] The sheet forging and thickening technique is applied to form the compressor attachment portion **100** in the chassis **70**. In this case, the fastening portion **115** is formed to have a thickness **T2** larger than a thickness of the chassis **70**, through a thickening process to secure a fastening torque between the fastening portion **115** and the bolt **80**. For example, in a case where a thickness **T1** of the support portion **110** is about 0.8 mm, the thickness **T2** of the fastening portion **115** may be about 1.15 mm to about 1.2 mm. In addition, in a case where the thickness **T1** of the support portion **110** is about 1.0 mm, the thickness **T2** of the fastening portion **115** may be about 1.2 mm to

about 1.25 mm. If the thickness T2 is less than about 1.0 mm, it may be difficult for the fastening portion **115** to secure the strength for supporting the bolt **80**.

[0067] The thickness T1 of the support portion **110** is equal to the thickness of other portions of the chassis **70**, such as the plate portion. For example, in a case where the thickness of the chassis **70** is about 0.8 mm, the thickness T1 of the support portion **110** may also be about 0.8 mm, and in a case where the thickness of the chassis **70** is about 1.0 mm, the thickness T1 of the support portion **110** may also be about 1.0 mm. The thickness T1 may represent a thickness of remaining portions of the chassis **70** excluding the fastening portion **115**, without being limited thereto. The thicknesses described herein may be taken in a direction normal to a major surface of the respective portion, such as in a direction perpendicular thereto. In an embodiment, the fastening portion **115**, the support portion **110** and the portion of the chassis **70** which is adjacent to the support portion **115** define a single body. A thickness of the support portion **110** may be the same as a thickness of the portion of the chassis which is adjacent to the support portion **1**.

[0068] FIG. **6** is a flow chart illustrating method of providing the compressor attachment portion **100** according to an embodiment of the disclosure. FIGS. **7A** to **7G** are cross-sectional diagrams illustrating the method of providing the compressor attachment portion **100** according to an embodiment of the disclosure.

[0069] Hereinafter, an example in which the compressor attachment portion **100** is formed in the chassis **70** by a sheet forging and thickening process according to an embodiment of the disclosure will be described with reference to the drawings. The chassis **70** used in the disclosure may have a thickness of about 0.8 mm to about 1.0 mm. In the following description, an example in which the chassis **70** has a thickness of about 0.8 mm will be used.

[0070] Referring to FIG. **7A**, a portion of the chassis **70** may be processed by the following process to integrally form the compressor attachment portion **100** with the chassis **70**. That is, the compressor attachment portion **100** and a remainder of the chassis **70** may together form a single body.

[0071] FIG. **7A** shows an area of the plate portion of the chassis **70** which corresponds to a location of the compressor attachment portion **100**. The area of the chassis **70** is disposed on a first lower die **210**. A first drawing punch **410** may be disposed in a guide hole **211** (e.g., a first guide hole) of the first lower die **210**. The guide hole **211** and a surrounding portion of the lower die **210** may together correspond to a dimension, shape, etc. of the compressor attachment portion **100** to be formed at the area of the chassis **70**. The first lower die **210** may be elastically supported at a lower portion, by an elastic member (not shown) so that the first lower die **210** is movable up and down. The elastic member may bias the first lower die **210** in the upward direction. The up-and-down direction may be the vertical direction of FIG. **7A**, that is, along a thickness direction of the chassis **70**.

[0072] The first lower die **210** may move down by being pressed by a first upper die **310** moving down. That is, the first lower die **210** and the first upper die **310** may move together with each other. When the first upper die **310** moves up and the pressing force applied to the first lower die **210** is released, the first lower die **210** can move up and return to its original position due to the elastic force of the elastic member. The first drawing punch **410** may be fixed not to move up and down. That is, the first lower die **210**, the first upper die **310** and the area of the chassis **70** may each move relative to the first drawing punch **410** since the first drawing punch **410** is fixed in position along the up-and-down direction. Therefore, when the first lower die **210** moves down, a leading end portion **411** (e.g., distal end) of the first drawing punch **410** may relatively protrude toward the upper surface of the first lower die **210** by a predetermined length. While the distal end of the first drawing punch **410** is within the guide hole **211**, the first lower die **210** may move along an outer surface of the first drawing punch **410** to protrude the distal end outside of the guide hole **211**.

[0073] A molding groove **311** (e.g., a first molding groove) corresponding to the guide hole **211** of

the first lower die **210** may be provided in the first upper die **310**. A cross-sectional shape or three-dimensional shape of the molding groove **311** of the first upper die **310** may correspond to the cross-sectional shape or three-dimensional shape of the leading end portion **411** of the first drawing punch **410**. For example, the leading end portion **411** of the first drawing punch **410** may be formed in a dome shape.

[0074] The first upper die **310** may face the first lower die **210** with an area of the chassis **70** therebetween. The molding method for providing the compressor attachment portion according to an embodiment of the disclosure may include a drawing process in which the first drawing punch **410** deforms a portion of the chassis **70** covering the guide hole **211** of the first lower die **210** (**601** in FIG. 6).

[0075] Referring to FIG. 7B, the first upper die **310** may move down to press the first lower die **210** downward, and the two dies may move in a same direction together with each other. In this case, the leading end portion **411** of the first drawing punch **410** may protrude from the guide hole **211** of the first lower die **210**. Here, a tip portion of the first drawing punch **410** may extend further than a top surface of the first lower die **210** to contact the chassis **70** which extends across the guide hole **211**.

[0076] A portion of the chassis **70** fixed between the first upper die **310** and the first lower die **210** may be deformed into a shape corresponding to the molding groove **311** of the first upper die **310** by being pressed toward the molding groove **311** of the first upper die **310** by the leading end portion **411** of the first drawing punch **410**. For example, a portion of the chassis **70** pressed by the leading end portion **411** of the first drawing punch **410** may be deformed from a flat shape to a curved shape.

[0077] The portion of the chassis **70** which is deformed to the curved shape (hereinafter referred to as a molded portion **111**) of the chassis **70** is a portion subjected to a drawing process (e.g., an initial drawing process) to form the fastening portion **115** of the compressor attachment portion **100**. Another portion of the chassis **70** which extends from and is adjacent to the molded portion **111** corresponds to the support portion **110** of the compressor attachment portion **100**. The molded portion **111** of FIG. 7B may represent a preliminary fastening portion.

[0078] The molding method for providing the compressor attachment portion **100** according to an embodiment of the disclosure may include first and second re-drawing processes for gradually increasing the height of the molded portion **111** of FIG. 7B and gradually decreasing the diameter of the molded portion **111** of FIG. 7B (**602** in FIG. 6). These re-drawing processes are preprocessing steps for increasing the thickness **T2** of the fastening portion **115**.

[0079] Referring to FIG. 7C, a second upper die **320** may move down to press a second lower die **220** downward. In this case, a leading end portion **421** of a second drawing punch **420** may protrude outwardly from a guide hole **221** (e.g., a second guide hole) of the second lower die **220**.

[0080] The molded portion **111** of the chassis **70** as a preliminary fastening portion which is fixed between the second upper die **320** and the second lower die **220** is subjected to a first re-drawing process in which the molded portion **111** of the chassis **70** is pressed toward a molding groove **321** (e.g., a second molding groove) of the second upper die **320** by the leading end portion **421** of the second drawing punch **420**.

[0081] The second drawing punch **420** may have a smaller diameter than the first drawing punch **410**. The molding groove **321** of the second upper die **320** may have a shape (e.g., cross-sectional or three-dimension) corresponding to the leading end portion **421** of the second drawing punch **420**. The molding groove **321** of the second upper die **320** may be formed to have a smaller diameter and a larger depth than the molding groove **311** of the first upper die **310**. Accordingly, the molded portion **111** can be deformed to have a greater depth and extend higher than the molded portion **111** illustrated in FIG. 7B, and the molded portion **111** can be deformed to be thinner than the molded portion **111** illustrated in FIG. 7B. The molded portion **111** which has the larger height and the smaller thickness may represent a first-drawn molded portion.

[0082] Referring to FIG. 7D, when a third upper die **330** moves down and presses a third lower die **230** downward, a leading end portion **431** of a third drawing punch **430** may protrude from a guide hole **231** (e.g., a third guide hole **231**) of the third lower die **230**.

[0083] The molded portion **111** of the chassis **70** which is first-drawn and fixed between the third upper die **330** and the third lower die **230** is subjected to a second re-drawing process in which the molded portion **111** of the chassis **70** is pressed toward a molding groove **331** (e.g., a third molding groove) of the third upper die **330** by the leading end portion **431** of the third drawing punch **430**.

[0084] The third drawing punch **430** may have a smaller diameter than the second drawing punch **420**. The molding groove **331** of the third upper die **330** may have a shape corresponding to the leading end portion **431** of the third drawing punch **430**. The molding groove **331** of the third upper die **330** may be formed to have a smaller diameter and a larger depth than the molding groove **321** of the second upper die **320**. Accordingly, the molded portion **111** which is first-drawn can be deformed to be higher than the molded portion **111** illustrated in FIG. 7C, and the molded portion **111** can be deformed to be thinner than the molded portion **111** illustrated in FIG. 7C. In this case, the thickness of the molded portion **111** may be about 0.5 mm. The molded portion **111** which has the height which is twice increased and the thickness which is twice decreased may represent a second-drawn molded portion.

[0085] Referring to FIG. 7E, the molding method for providing the compressor attachment portion according to an embodiment of the disclosure may include a piercing process for forming a hole at an upper end (e.g., a distal end) of the molded portion **111** (**603** in FIG. 6). The process of forming a through hole **113** of the molded portion **111** may be a preprocessing process capable of smoothly performing a burring process.

[0086] A molding groove **341** (e.g., a fourth molding groove) of a fourth upper die **340** may have the same shape and size as the molding groove **331** of the third upper die **330**. The fourth upper die **340** may have a guide hole **342** as a punch guide hole defined therein, which is open at opposing surfaces of the die.

[0087] A fourth lower die **240** may have a support protrusion **241** corresponding to the molding groove **341** of the fourth upper die **340**. The support protrusion **241** may be a part of the fourth lower die **240**, such as a protruded portion thereof.

[0088] The support protrusion **241** of the fourth lower die **240** may be inserted into the molded portion **111** to support the molded portion **111**.

[0089] A first piercing punch **510** may be extended through the guide hole **342** provided in the fourth upper die **340** to perforate the center of the upper end of the molded portion **111**. The through hole **113** may be formed at the center of the upper end of the molded portion **111** by the first piercing punch **510**. The through hole **113** of the molded portion **111** may have a diameter approximately corresponding to the diameter of the first piercing punch **510**. The molded portion **111** which has the through hole **113** defined therein may represent a pierced molded portion.

[0090] The molding method for providing the compressor attachment portion according to an embodiment of the disclosure may include a burring process and a first forging process as preprocessing steps for molding the molded portion **111** (e.g., the pierced molded portion) into in the final form of the fastening portion **115** (**604** in FIG. 6).

[0091] Referring to FIG. 7F, when a fifth upper die **350** moves down to press a fifth lower die **250** in a state where the molded portion **111** of the chassis **70** which is pierced is disposed on the fifth lower die **250**, portions of the chassis **70** extended from the through hole **113** move downward together with the fifth lower die **250** and the fifth upper die **350**. In this case, the portions of the chassis **70** which form the pierced molded portion may be deformed into a burring shape by a first burring punch **710** penetrating through the chassis **70** at the through hole **113**, where the first burring punch **710** is fixed not to move up and down. The first burring punch **710** may be inserted into a guide hole **251** (e.g., a fifth guide hole) of the fifth lower die **250**.

[0092] In an embodiment, the portions of the chassis **70** extended from the through hole **113**, the

fifth lower die **250** and the fifth upper die **350** move together with each other along an outer surface of the first burring punch **710** to widen the through hole **113** and separate the portions of the chassis **70** which are extended therefrom. The molded portion **111** which has the through hole **113** widened may represent a burred molded portion. The portions of the chassis **70** in the burred molded portion may be a preliminary form of the fastening portion **115**.

[0093] An outer circumferential surface of the molded portion **111** deformed into the burring shape may come into contact with a guide hole **351** (e.g., a fifth guide hole) of the fifth upper die **350**, and an inner circumferential surface of the molded portion **111** may come into contact with an outer surface of the first burring punch **710**. In this state, a first forging punch **610** moves up and down along the guide hole **351** of the fifth upper die **350**, striking the upper end of the molded portion **111** to increase the thickness of the molded portion **111**. Here, a downward force of the first forging punch **610** compresses the portions of the chassis **70** in a thickness direction (e.g., vertical in FIG. 7F) to increase a thickness of the burred molded portion. The molded portion **111** which has the increased thickness (and decreased height) may represent a first-forged molded portion.

[0094] In FIGS. 7B through 7F, a thickness of the support portion **110** may remain substantially the same.

[0095] The molding method for providing the compressor attachment portion according to an embodiment of the disclosure may include a second forging process to increase the thickness of the first-forged molded portion **111** (**605** in FIG. 6).

[0096] Referring to FIG. 7G, the chassis **70** is disposed on a sixth lower die **260**. In this case, a support protrusion **261** formed at an upper surface of the sixth lower die **260** may be inserted into the first-forged molded portion **111**.

[0097] In this state, when a sixth upper die **360** moves down, the molded portion **111** is inserted into a guide hole **361** (e.g., a sixth guide hole) of the sixth upper die **360**. A second forging punch **620** moves up and down along the guide hole **361** of the sixth upper die **360**, striking the upper end of the molded portion **111** to further increase the thickness of the molded portion **111**. Accordingly, the thickness of the molded portion **111** can be formed to be thicker than the thickness of the support portion **110** and/or a remainder of the chassis **70**. For example, by the secondary forging process, the thickness of the molded portion **111** can be increased to about 1.15 mm to about 2.0 mm, and the height of the molded portion **111** can be about 4.0 mm to about 4.5 mm. The molded portion **111** which has the further increased thickness (and further decreased height) may represent a second-forged molded portion.

[0098] After the second forging process is completed, a tapping process (not shown) may be performed on the inner circumferential surface of the molded portion **111** to form a screw thread. The second-forged molded portion having the screw thread defined therein may substantially represent the compressor attachment portion **100** having the fastening portion **115** with a larger thickness than a remainder of the chassis **70**.

[0099] Although not shown, between processes shown in FIGS. 7A through 7G and between FIG. 7G and the tapping process, upper and lower dies and corresponding punches, support protrusions, etc. of a previous process may be separated from the area of the chassis **70** and upper and lower dies and corresponding punches, support protrusions, etc. of a next process may be disposed relative to the area of the chassis **70** as shown in the cross-sectional views.

[0100] The above-described forming method can be applied when the thickness of the chassis **70** is about 0.8 mm to about 1.0 mm. In a case where a chassis **70'** (see FIG. 9A) has a thickness of more than about 1.0 mm (for example, about 1.2 mm to about 1.4 mm), the compressor attachment portion can be integrally formed with the chassis **70'** by a molding method other than the above-described molding method. In this case, when the thickness of the support portion is about 1.2 mm to about 1.4 mm, the thickness of the fastening portion can be approximately 1.1 mm to approximately 1.3 mm.

[0101] Hereinafter, a process for integrally forming a compressor attachment portion with the

chassis **70'** as a single body (see FIG. **9A**) when the thickness of the chassis **70'** is about 1.4 mm will be described with reference to the drawings. Descriptions above with details about processes in FIGS. **7A** to **7G** may be applied to similar processes described below for FIGS. **9A-9H**.

[0102] In an embodiment, a method for providing a compressor chassis (e.g., chassis **70**) of a cooling appliance includes providing a chassis as a single body defining a fastening portion **115** at which a compressor **51** of a cool air generator is fastened to the chassis **70** by a fastening member screw-coupled to the chassis **70**, and a support portion **110** which is extended from the fastening portion **115**, has a thickness and connects the fastening portion **115** to a portion of the chassis which is adjacent to the support portion. The method includes in a drawing process (FIGS. **7A** and **7B**), deforming a portion of the chassis **70** which corresponds to the fastening portion **115** using a first drawing punch **410** having a diameter, to provide a molded portion **111** of the chassis **70** which is convex and has a height, a thickness and a diameter. The method includes in a re-drawing process (FIGS. **7C** and **7D**), increasing the height, decreasing the thickness and decreasing the diameter of the molded portion **111** using a second drawing punch having a diameter which is smaller than the diameter of the first drawing punch, to provide a re-drawn molded portion of the chassis having a height, a thickness and a diameter. The method includes in a piercing process (FIG. **7E**), perforating an upper end of the re-drawn molded portion using a piercing punch to provide a through hole penetrating the upper end of the re-drawn molded portion. The method includes in a first forging process (FIG. **7F**), widening the through hole of the re-drawn molded portion by penetrating the through hole using a burring punch, and increasing the thickness of the re-drawn molded portion using a first forging punch striking an upper end of the re-drawn molded portion, to provide the fastening portion having a first thickness. The method includes in a second forging process (FIG. **7F**), striking an upper end of the fastening portion which has the first thickness using a second forging punch to provide the fastening member having a second thickness which is larger than the first thickness and larger than the thickness of the support portion.

[0103] The re-drawing process may include a first re-drawing process (FIG. **7C**) which deforms the molded portion to provide the molded portion having a first height and a first diameter, and a second re-drawing process (FIG. **7D**) which deforms the molded portion which has the first height and the first diameter to provide the molded portion having a second height larger than the first height and a second diameter smaller than the first diameter.

[0104] The method may further include providing a screw thread on an inner circumferential surface of the fastening portion having the second thickness at which the fastening member is screw-coupled to the chassis.

[0105] FIG. **8** is a flow chart illustrating a method of providing the compressor attachment portion **100** according to an embodiment of the disclosure. FIGS. **9A** to **9H** are cross-sectional diagrams illustrating the method of providing the compressor attachment portion according to an embodiment of the disclosure.

[0106] Referring to FIG. **9A**, the chassis **70'** is disposed on a seventh lower die **1210**. A third drawing punch **1410** may be disposed in a guide hole **1211** of the seventh lower die **1210**. A lower portion of the seventh lower die **1210** may be elastically supported by an elastic member so that the seventh lower die **1210** is movable up and down.

[0107] The seventh lower die **1210** may move down by being pressed by a seventh upper die **1310** moving down. When the seventh upper die **1310** moves up and the pressing force applied to the seventh lower die **1210** is released, the seventh lower die **1210** can move up and return to its original position due to the elastic force (e.g., a biasing force) of the elastic member. A fourth drawing punch **1410** may be fixed not to move up and down, but dies may be moveable relative thereto. Therefore, when the seventh lower die **1210** moves down, a leading end portion **1411** of the fourth drawing punch **1410** may relatively protrude toward the upper surface of the seventh lower die **1210** by a predetermined length.

[0108] A molding groove **1311** corresponding to the guide hole **1211** of the seventh lower die **1210**

may be provided in the seventh upper die **1310**. The molding groove **1311** of the seventh upper die **1310** may correspond to the shape of the leading end portion **1411** of the fourth drawing punch **1410**. For example, the leading end portion **1411** of the fourth drawing punch **1410** may be formed in a dome shape.

[0109] The molding method for providing the compressor attachment portion **100** according to an embodiment of the disclosure may include a drawing process in which the fourth drawing punch **1410** deforms a portion of the chassis **70'** covering the guide hole **1211** of the seventh lower die **1210** (**801** in FIG. 8).

[0110] Referring to FIG. 9B, the seventh upper die **1310** may move down to press the seventh lower die **1210** downward. In this case, the leading end portion **1411** of the fourth drawing punch **1410** may protrude from the guide hole **1211** of the seventh lower die **1210**.

[0111] A portion of the chassis **70'** fixed between the seventh upper die **1310** and the seventh lower die **1210** may be deformed into a shape corresponding to the molding groove **1311** of the seventh upper die **1310** by being pressed toward the molding groove **1311** of the seventh upper die **1310** by the leading end portion **1411** of the fourth drawing punch **1410**. For example, a portion of the chassis **70'** pressed by the leading end portion **1411** of the fourth drawing punch **1410** may be deformed from a flat shape to a curved shape.

[0112] A molded portion **1111** deformed to the curved shape of the chassis **70'** is a portion subjected to a drawing process to form a fastening portion **115** of the compressor attachment portion **100**. Another portion of the chassis **70'** adjacent to the molded portion **1111** corresponds to a support portion **1110** of the compressor attachment portion.

[0113] The molding method for providing the compressor attachment portion according to an embodiment of the disclosure may include a piercing process for forming a hole at an upper end of the molded portion **1111** (**802** in FIG. 8).

[0114] Referring to FIG. 9C, a molding groove **1321** of an eighth upper die **1320** may have the same shape and size as the molding groove **1311** of the seventh upper die **1310**. An eighth lower die **1220** may have a support protrusion **1221** corresponding to the molding groove **1321** of the eighth upper die **1320**.

[0115] The support protrusion **1221** of the eighth lower die **1220** may be inserted into the molded portion **1111** to support the molded portion **1111**.

[0116] A second piercing punch **1510** may be brought into a guide hole **1322** provided in the eighth upper die **1320** to perforate the center of the upper end of the molded portion **1111**. A through hole **1113** may be formed at the center of the upper end of the molded portion **1111** by the second piercing punch **1510**. The through hole **1113** of the molded portion **1111** may have a diameter approximately corresponding to the diameter of the second piercing punch **1510**.

[0117] The molding method for providing the compressor attachment portion **100** according to an embodiment of the disclosure may include first, second, and third re-drawing processes for gradually increasing the height of the molded portion **1111** of FIG. 9C and gradually decreasing the diameter of the molded portion **1111** of FIG. 9C (**803** in FIG. 8). The plurality of re-drawing processes are preprocessing steps for increasing the thickness of the fastening portion **115** of the compressor attachment portion **100**. The re-drawing process may be performed more times in this molding method for providing the compressor attachment portion **100** than in the above-described molding method for the compressor attachment portion **100**, considering that the chassis **70'** has a larger thickness than the chassis **70** described above.

[0118] Referring to FIG. 9D, a ninth upper die **1330** may move down to press a ninth lower die **1230** downward. In this case, a leading end portion **1431** of a fifth drawing punch **1430** may protrude from a guide hole **1231** of the ninth lower die **1230**.

[0119] The molded portion **1111** of the chassis **70'** fixed between the ninth lower die **1230** and the ninth upper die **1330** is subjected to a first re-drawing process in which the molded portion **1111** of the chassis **70'** is pressed toward a molding groove **1331** of the ninth upper die **1330** by the leading

end portion **1431** of the fifth drawing punch **1430**.

[0120] The molding groove **1331** of the ninth upper die **1330** may be formed to have a smaller diameter and a larger depth than the molding groove **1321** of the eighth upper die **1320**.

Accordingly, the molded portion **1111** having the through hole **1113** defined therein can be deformed to be higher than the molded portion **1111** illustrated in FIG. 9B, and the molded portion **1111** can be deformed to be thinner than the molded portion **1111** illustrated in FIG. 9B. The molded portion **1111** after the process in FIG. 9D may be a first-drawn molded portion.

[0121] Referring to FIG. 9E, a tenth upper die **1340** may move down to press a tenth lower die **1240** downward. In this case, a leading end portion **1441** of a sixth drawing punch **1440** may protrude from a guide hole **1241** of the tenth lower die **1240**.

[0122] The molded portion **1111** of the chassis **70'** fixed between the tenth lower die **1240** and the tenth upper die **1340** is subjected to a second re-drawing process in which the molded portion **1111** of the chassis **70'** which is first-drawn is pressed toward a molding groove **1341** of the tenth upper die **1340** by the leading end portion **1441** of the sixth drawing punch **1440**.

[0123] The molding groove **1341** of the tenth upper die **1340** may be formed to have a smaller diameter and a larger depth than the molding groove **1331** of the ninth upper die **1330**.

Accordingly, the molded portion **1111** can be deformed to be higher than the molded portion **1111** illustrated in FIG. 9D, and the molded portion **1111** can be deformed to be thinner than the molded portion **1111** illustrated in FIG. 9D. In addition, the through hole **1113** formed at the center of the upper end of the molded portion **1111** may have a larger size than that in the molded portion **1111** illustrated in FIG. 9D. The molded portion **1111** after the process in FIG. 9E may be a second-drawn molded portion.

[0124] Referring to FIG. 9F, an eleventh upper die **1350** may move down to press an eleventh lower die **1250** downward. In this case, a leading end portion **1451** of a seventh drawing punch **1450** may protrude from a guide hole **1251** of the eleventh lower die **1250**.

[0125] The molded portion **1111** of the chassis **70'** fixed between the eleventh lower die **1250** and the eleventh upper die **1350** is subjected to a third re-drawing process in which the molded portion **1111** of the chassis **70'** is pressed toward a molding groove **1351** of the eleventh upper die **1350** by a leading end portion **1451** of a seventh drawing punch **1450**.

[0126] The molding groove **1351** of the eleventh upper die **1350** may be formed to have a smaller diameter and a larger depth than the molding groove **1341** of the tenth upper die **1340**. Accordingly, the molded portion **1111** can be deformed to be higher than the molded portion **1111** illustrated in FIG. 9E, and the molded portion **1111** can be deformed to be thinner than the molded portion **1111** illustrated in FIG. 9E. In addition, the through hole **1113** formed at the center of the upper end of the molded portion **1111** may have a larger size than that in the molded portion **1111** illustrated in FIG. 9E. The molded portion **1111** after the process in FIG. 9F may be a third-drawn molded portion.

[0127] The molding method for providing the compressor attachment portion **100** according to an embodiment of the disclosure may include a molding process for gradually increasing the height of the molded portion **1111** of FIG. 9F and gradually decreasing the diameter of the molded portion **1111** of FIG. 9F (**804** in FIG. 8).

[0128] Referring to FIG. 9G, a twelfth upper die **1360** may move down to press a twelfth lower die **1260** downward. In this case, a leading end portion **1611** of a molding punch **1610** may protrude from a guide hole **1261** of the twelfth lower die **1260**.

[0129] The molded portion **1111** of the chassis **70'** fixed between the twelfth lower die **1260** and the twelfth upper die **1360** is subjected to a molding process in which the molded portion **1111** of the chassis **70'** is pressed toward a molding groove **1361** of the twelfth upper die **1360** by the leading end portion **1611** of the molding punch **1610**.

[0130] Through the molding process, the molded portion **1111** can be deformed to have a smaller diameter than the molded portion **1111** illustrated in FIG. 9F. However, the height of the molded

portion **1111** can be kept substantially equal to the height of the molded portion **1111** illustrated in FIG. 9F.

[0131] The molding method for providing the compressor attachment portion **100** according to an embodiment of the disclosure may include a burring process for molding the molded portion **1111** in the preliminary form shape of the fastening portion **115** of the compressor attachment portion **100** (**805** in FIG. 8).

[0132] Referring to FIG. 9H, when a thirteenth upper die **1370** moves down to press a thirteenth lower die **1270** in a state where the molded portion **1111** of the chassis **70'** is disposed on the thirteenth lower die **1270**, the chassis **70'** moves downward together with the thirteenth lower die **1270**.

[0133] In this case, an upper portion of the molded portion **1111** may be penetrated by a second burring punch **1710** fixed not to move up and down. After penetrating the upper portion of the molded portion **1111**, the second burring punch **1710** is brought into a guide hole **1371** of the thirteenth upper die **1370**. Through the burring process, the molded portion **1111** can be deformed into a burring shape and molded in the form of the fastening portion **115** of the compressor attachment portion **100**.

[0134] After the burring process is completed, a screw thread may be formed on the inner circumferential surface of the molded portion **1111** through a tapping process on the molded portion **1111**.

[0135] In an embodiment, a method for providing a compressor chassis of a cooling appliance, the method includes in a drawing process (FIGS. 9A and 9B), deforming a portion of the chassis which corresponds to the fastening portion using a first drawing punch having a diameter, to provide a molded portion of the chassis which is convex and has a height, a thickness and a diameter. The method includes in a piercing process (FIG. 9C), perforating an upper end of the molded portion using a piercing punch to provide a through hole penetrating the upper end of the molded portion. The method includes in a re-drawing process (FIGS. 9D, 9E and 9F), increasing the height, decreasing the thickness and decreasing the diameter of the molded portion having the through hole using a second drawing punch having a diameter which is smaller than the diameter of the first drawing punch, to provide a re-drawn molded portion of the chassis having a height, a thickness and a diameter. The method includes in a molding process (e.g., FIG. 9G), decreasing the diameter of the re-drawn molded portion using a molding punch having a smaller diameter than the diameter of the second drawing punch. The method includes in a forging process (FIG. 9H, also referred to as a burring process), widening the through hole of the re-drawn molded portion by penetrating the through hole using a burring punch to provide the fastening portion having a thickness smaller than the thickness of the support portion.

[0136] The re-drawing process may be repeated including a plurality of re-drawing processes each of which respectively increases a previous thickness together with decreasing a previous diameter of the molded portion (FIGS. 9D, 9E and 9F, in order, for example).

[0137] The fastening portion **115** of the compressor attachment portion **100** formed according to the above-described molding method for the compressor attachment portion **100** may have a thickness of about 1.3 mm, which is smaller than the thickness of the support portion **1110** of the compressor attachment portion **100**, which is about 1.4 mm. However, even though the thickness of walls of the fastening portion **115** are smaller than the thickness of the support portion **1110** and/or remaining portions of the chassis **70'** at the compressor attachment portion **100**, since the thickness of the fastening portion **115** of the compressor attachment portion **100** is about 1.0 mm or more, it is possible to satisfy the fastening torque required between the fastening portion **115** and the bolt **80**. That is, one or more embodiment of a method of providing a fastening portion **115** may form a thickness of the fastening portion **115** as a minimum of about 1.0 mm or more.

[0138] In addition, in a case where the chassis **70'** has a thickness of about 1.2 mm, the compressor attachment portion **100** can be integrally formed with the chassis **70'** as a single body through the

above-described molding method. In this case, the fastening portion **115** of the compressor attachment portion **100** may have a thickness of about 1.1 mm, which is smaller than the thickness of the support portion **1110** of the compressor attachment portion **100**, which is about 1.2 mm. However, since the thickness of the fastening portion **115** of the compressor attachment portion **100** is about 1.0 mm or more, it is possible to satisfy the fastening torque required between the fastening portion **115** and the bolt **80**.

[0139] The compressor attachment portion **100** according to one or more embodiment of the disclosure does not require a separate bracket to connect the chassis **70** and the compressor **51** to each other. In addition, since the compressor attachment portion **100** according to an embodiment of the disclosure is formed integrally with the chassis **70** to provide a single body by processing a partial area of the chassis **70**, a process of spot-welding a separate bracket to the chassis **70** can be omitted, thereby reducing the number of parts and assembly work, as a result improving productivity and reducing manufacturing costs.

[0140] In an embodiment, a cooling appliance include a main body in which a compressor chamber is defined, a chassis which is in the compressor chamber, the chassis being integrally formed and defining a fastening portion to which a fastening member is screw-coupled to the chassis, and a support portion which is extended from the fastening portion and connects the fastening portion to a portion of the chassis which is adjacent to the support portion (like at upper surface **71** of the chassis **70**, and a compressor of a cool air generator which is in the compressor chamber, penetrated by the fastening member and detachably attached to the chassis by the fastening member.

[0141] In the cooling appliance, the fastening portion and the support portion together may define a single body, a through hole may be defined in the chassis at the fastening portion, the fastening member being screw-coupled to the chassis via the through hole, and the fastening portion may include a drawn-and-pierced portion of the single body. Here, the drawn-and-pierced portion of the single body may define the through hole and a thickness of the chassis at the fastening portion which is larger (or smaller) than a thickness of the chassis at the support portion, according to embodiments of the methods disclosed herein.

[0142] In the cooling appliance, an upper surface of the support portion may be recessed from the portion of the chassis which is adjacent to the support portion (FIG. **4**, for example), and the fastening portion may protrude from the upper surface of the support portion and includes the drawn-and-pierced portion of the single body.

[0143] Although embodiments of the disclosure have been illustrated and described above, the disclosure is not limited to the specific embodiments described above, and various modification may be made by those skilled in the art without departing from the gist of the disclosure as claimed in the claims. Such modifications should not be individually understood from the technical spirit or prospect of the disclosure.

Claims

1. A cooling appliance comprising: a main body in which a compressor chamber is defined; a chassis which is in the compressor chamber, the chassis defining: a fastening portion to which a fastening member is screw-coupled to the chassis, and a support portion which is extended from the fastening portion and connects the fastening portion to a portion of the chassis which is adjacent to the support portion; and a compressor of a cool air generator which is in the compressor chamber, penetrated by the fastening member and detachably attached to the chassis by the fastening member.
2. The cooling appliance as claimed in claim 1, wherein the fastening portion, the support portion and the portion of the chassis which is adjacent to the support portion define a single body, and a thickness of the support portion is the same as a thickness of the portion of the chassis which is

adjacent to the support portion.

3. The cooling appliance as claimed in claim 2, wherein within the single body of the chassis, the thickness of the support portion is about 0.8 millimeter to about 1.0 millimeter.

4. The cooling appliance as claimed in claim 3, wherein within the single body of the chassis, a thickness of the fastening portion is larger than the thickness of the support portion.

5. The cooling appliance as claimed in claim 4, wherein the thickness of the fastening portion is about 1.15 millimeters to about 1.25 millimeters.

6. The cooling appliance as claimed in claim 2, wherein the thickness of the support portion is about 1.2 millimeters to about 1.4 millimeters.

7. The cooling appliance as claimed in claim 6, wherein a thickness of the fastening portion is smaller than the thickness of the support portion.

8. The cooling appliance as claimed in claim 7, wherein the thickness of the fastening portion is about 1.1 millimeters to about 1.3 millimeters.

9. The cooling appliance as claimed in claim 1, wherein the fastening portion, the support portion and the portion of the chassis which is adjacent to the support portion define a single body, an upper surface of the support portion faces the compressor, and the fastening portion protrudes from the upper surface and has a height of about 4.0 millimeters to about 4.5 millimeters from the upper surface of the support portion.

10. A method for providing a compressor chassis of a cooling appliance, the method comprising: providing a chassis as a single body defining: a fastening portion at which a compressor of a cool air generator is fastened to the chassis by a fastening member screw-coupled to the chassis, and a support portion which is extended from the fastening portion, has a thickness and connects the fastening portion to a portion of the chassis which is adjacent to the support portion; in a drawing process, deforming a portion of the chassis which corresponds to the fastening portion using a first drawing punch having a diameter, to provide a molded portion of the chassis which is convex and has a height, a thickness and a diameter; in a re-drawing process, increasing the height, decreasing the thickness and decreasing the diameter of the molded portion using a second drawing punch having a diameter which is smaller than the diameter of the first drawing punch, to provide a re-drawn molded portion of the chassis having a height, a thickness and a diameter; in a piercing process, perforating an upper end of the re-drawn molded portion using a piercing punch to provide a through hole penetrating the upper end of the re-drawn molded portion; in a first forging process: widening the through hole of the re-drawn molded portion by penetrating the through hole using a burring punch, and increasing the thickness of the re-drawn molded portion using a first forging punch striking an upper end of the re-drawn molded portion, to provide the fastening portion having a first thickness; and in a second forging process, striking an upper end of the fastening portion which has the first thickness using a second forging punch to provide the fastening member having a second thickness which is larger than the first thickness and larger than the thickness of the support portion.

11. The method as claimed in claim 10, wherein the re-drawing process includes: a first re-drawing process which deforms the molded portion to provide the molded portion having a first height and a first diameter; and a second re-drawing process which deforms the molded portion which has the first height and the first diameter to provide the molded portion having a second height larger than the first height and a second diameter smaller than the first diameter.

12. The method as claimed in claim 11, further comprising providing a screw thread on an inner circumferential surface of the fastening portion having the second thickness at which the fastening member is screw-coupled to the chassis.

13. A method for providing a compressor chassis of a cooling appliance, the method comprising: providing a chassis as a single body defining: a fastening portion at which a compressor of a cool air generator is fastened to the chassis by a fastening member screw-coupled to the chassis, and a support portion which is extended from the fastening portion, has a thickness and connects the

fastening portion to a portion of the chassis which is adjacent to the support portion; in a drawing process, deforming a portion of the chassis which corresponds to the fastening portion using a first drawing punch having a diameter, to provide a molded portion of the chassis which is convex and has a height, a thickness and a diameter; in a piercing process, perforating an upper end of the molded portion using a piercing punch to provide a through hole penetrating the upper end of the molded portion; in a re-drawing process, increasing the height, decreasing the thickness and decreasing the diameter of the molded portion having the through hole using a second drawing punch having a diameter which is smaller than the diameter of the first drawing punch, to provide a re-drawn molded portion of the chassis having a height, a thickness and a diameter; in a molding process, decreasing the diameter of the re-drawn molded portion using a molding punch having a smaller diameter than the diameter of the second drawing punch; and in a forging process, widening the through hole of the re-drawn molded portion by penetrating the through hole using a burring punch to provide the fastening portion having a thickness smaller than the thickness of the support portion.

14. The method as claimed in claim 13, wherein the re-drawing process is repeated including a plurality of re-drawing processes each of which respectively increases a previous thickness together with decreasing a previous diameter of the molded portion.

15. The method as claimed in claim 13, further comprising providing a screw thread on an inner circumferential surface of the fastening portion having the second thickness at which the fastening member is screw-coupled to the chassis.

16. The cooling appliance as claimed in claim 1, wherein the fastening portion and the support portion together define a single body, a through hole is defined in the chassis at the fastening portion, the fastening member being screw-coupled to the chassis via the through hole, and the fastening portion includes a drawn-and-pierced portion of the single body.

17. The cooling appliance as claimed in claim 16, wherein the drawn-and-pierced portion of the single body defines the through hole and a thickness of the chassis at the fastening portion which is larger than a thickness of the chassis at the support portion.

18. The cooling appliance as claimed in claim 16, wherein the drawn-and-pierced portion of the single body defines the through hole and a thickness of the chassis at the fastening portion which is smaller than a thickness of the chassis at the support portion.

19. The cooling appliance as claimed in claim 16, wherein an upper surface of the support portion is recessed from the portion of the chassis which is adjacent to the support portion, and the fastening portion protrudes from the upper surface of the support portion and includes the drawn-and-pierced portion of the single body.
